

Bibliography

References

- [1] J. Adámek, H. Herrlich, and G. E. Strecker, *Abstract and Concrete Categories: The Joy of Cats*. Wiley, 1990.
- [2] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed. Springer, 1998.
- [3] S. Eilenberg, *Automata, Languages, and Machines*, Vol. A. Academic Press, 1974.
- [4] E. L. Post, “The Two-Valued Iterative Systems of Mathematical Logic,” *Annals of Mathematics Studies*, no. 5, Princeton University Press, 1941.
- [5] E. Börger, E. Grädel, and Y. Gurevich, *The Classical Decision Problem*. Springer, 2001.
- [6] P. Jeavons, D. Cohen, and M. Gyssens, “Closure properties of constraints,” *Journal of the ACM*, vol. 44, no. 4, pp. 527–548, 1997.
- [7] S. Burris and H. P. Sankappanavar, *A Course in Universal Algebra*. Springer, 1981.
- [8] F. Bergeron, G. Labelle, and P. Leroux, *Combinatorial Species and Tree-like Structures*. Cambridge University Press, 1998.
- [9] A. Joyal, “Une théorie combinatoire des séries formelles,” *Advances in Mathematics*, vol. 42, no. 1, pp. 1–82, 1981.
- [10] R. Street, “Quantum groups: a path to current algebra,” in *The Proceedings of the 1986 AMS Summer Research Institute*, 2007 reprint and related categorical foundations.
- [11] S. Lack, “A 2-categories companion,” in *Towards Higher Categories*, Springer, 2010.
- [12] P. Selinger, “A survey of graphical languages for monoidal categories,” in *New Structures for Physics*, Springer, 2011, pp. 289–355.
- [13] B. Coecke and A. Kissinger, *Picturing Quantum Processes*. Cambridge University Press, 2017.
- [14] P. Selinger, “Towards a quantum programming language,” *Mathematical Structures in Computer Science*, vol. 14, no. 4, pp. 527–586, 2004.
- [15] T. Hales, “Formal proof,” *Notices of the AMS*, vol. 55, no. 11, pp. 1370–1380, 2008.
- [16] G. Gonthier, “Formal proof—the four-color theorem,” *Notices of the AMS*, vol. 55, no. 11, pp. 1382–1393, 2008.
- [17] D. E. Knuth, *The Art of Computer Programming, Vol. 1: Fundamental Algorithms*, 3rd ed. Addison-Wesley, 1997.
- [18] M. Huth and M. Ryan, *Logic in Computer Science: Modelling and Reasoning about Systems*, 2nd ed. Cambridge University Press, 2004.
- [19] B. C. Pierce, *Types and Programming Languages*. MIT Press, 2002.
- [20] J.-Y. Girard, Y. Lafont, and P. Taylor, *Proofs and Types*. Cambridge University Press, 1989.

- [21] P. T. Johnstone, *Stone Spaces*. Cambridge University Press, 1982.
- [22] G. Birkhoff, “Lattice theory,” *American Mathematical Society Colloquium Publications*, 1940.
- [23] B. D. McKay, “Practical graph isomorphism,” *Congressus Numerantium*, vol. 30, pp. 45–87, 1981.
- [24] L. Babai, “Graph isomorphism in quasipolynomial time,” in *Proceedings of the 48th ACM Symposium on Theory of Computing*, 2016, pp. 684–697.
- [25] R. Diestel, *Graph Theory*, 5th ed. Springer, 2017.
- [26] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley, 2006.
- [27] C. E. Shannon and W. Weaver, *The Mathematical Theory of Communication*. University of Illinois Press, 1949.
- [28] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*. Cambridge University Press, 2000.
- [29] O. Goldreich, *Foundations of Cryptography*, Vol. 1. Cambridge University Press, 2001.
- [30] D. R. Stinson, *Cryptography: Theory and Practice*, 3rd ed. Chapman & Hall/CRC, 2005.
- [31] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.
- [32] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković, “Geometric deep learning: Grids, groups, graphs, geodesics, and gauges,” *arXiv:2104.13478*, 2021.
- [33] R. Wille and R. Drechsler, *Towards a Design Flow for Reversible Logic*. Springer, 2009.
- [34] M. Amy, D. Maslov, M. Mosca, and M. Roetteler, “A meet-in-the-middle algorithm for fast synthesis of depth-optimal quantum circuits,” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 32, no. 6, pp. 818–830, 2013.
- [35] The Agda Team, *Agda Documentation*. Available as project documentation; accessed for formalization patterns.
- [36] The Coq Development Team, *The Coq Proof Assistant Reference Manual*. Available as project documentation; accessed for proof engineering patterns.